Domain of Rational Functions

Algebra Worksheet · Grade 9–11

Name: _____ Date: _____

Learning Objectives

- Identify rational functions as ratios of two polynomials where the denominator is not zero
- Find values of x that make the denominator equal to zero
- Express the domain of a rational function using set notation and words

Problems

1. Find the domain of the rational function below. Write your answer in words.

$$f(x) = \frac{5}{x-3}$$

2. Find the domain of the rational function below. Write your answer in words.

$$f(x) = \frac{4x}{x+7}$$

3. Find the domain of the rational function below. Write your answer using set notation.

$$g(x) = \frac{2x+1}{x-5}$$

4. Find the domain of the rational function below. Write your answer in words.

$$h(x) = \frac{3}{x^2-9}$$

5. Find the domain of the rational function below. Write your answer in set notation.

$$f(x) = \frac{4}{x^2+1}$$

6. Find the domain of the rational function below. Write your answer in set notation.

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$$p(x) = \frac{3x}{x^2 - 10x - 24}$$

7. Find the domain of the rational function below. Write your answer in words.

$$f(x) = \frac{2}{(x-1)^2}$$

8. Find the domain of the rational function below. Write your answer in set notation.

$$f(x) = \frac{x^2 + 3x}{x^2 - 16}$$

9. Find the domain of the rational function below. Write your answer in set notation.

$$f(x) = \frac{5x - 2}{x^2 + 3x - 18}$$

10. Find the domain of the rational function below. Then explain in one sentence why the excluded values cannot be in the domain.

$$f(x) = \frac{x^2 - 4}{x^3 - 4x^2 - 7x + 10}$$

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Domain of Rational Functions — Answer Key

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Answer Key

1. Answer: All real numbers except x = 3

- Set the denominator equal to zero: x - 3 = 0
- Solve for x: x = 3
- The domain is all real numbers except x = 3

2. Answer: All real numbers except x = -7

- Set the denominator equal to zero: x + 7 = 0
- Solve for x: x = -7
- The domain is all real numbers except x = -7

3. Answer: { x ∈ R | x ≠ 5 }

- Set the denominator equal to zero: x - 5 = 0
- Solve: x = 5
- Write in set notation: { x ∈ **R** | x ≠ 5 }

4. Answer: All real numbers except x = 3 and x = -3

- Set the denominator equal to zero: $x^2 - 9 = 0$
- Factor as a difference of two squares: (x - 3)(x + 3) = 0
- Apply the zero product property: x = 3 or x = -3
- The domain is all real numbers except x = ±3

5. Answer: { x ∈ R } (all real numbers)

- Set the denominator equal to zero: $x^2 + 1 = 0$
- Solve: $x^2 = -1$
- There is no real solution since a square cannot be negative
- The denominator is never zero, so the domain is all real numbers

6. Answer: { x ∈ R | x ≠ 12 and x ≠ -2 }

- Set the denominator equal to zero: $x^2 - 10x - 24 = 0$
- Factor the trinomial: (x - 12)(x + 2) = 0
- Apply zero product property: x = 12 or x = -2
- Domain: { x ∈ **R** | x ≠ 12 and x ≠ -2 }

7. Answer: All real numbers except x = 1

- Set the denominator equal to zero: $(x - 1)^2 = 0$
- Take the square root of both sides: x - 1 = 0
- Solve: x = 1
- The domain is all real numbers except x = 1

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8. Answer: { x ∈ R | x ≠ 4 and x ≠ -4 }

- Set the denominator equal to zero: $x^2 - 16 = 0$
- Factor as difference of two squares: $(x - 4)(x + 4) = 0$
- Apply zero product property: $x = 4$ or $x = -4$
- Domain: { x ∈ **R** | x ≠ ±4 }

9. Answer: { x ∈ R | x ≠ 3 and x ≠ -6 }

- Set the denominator equal to zero: $x^2 + 3x - 18 = 0$
- Factor the trinomial: $(x + 6)(x - 3) = 0$
- Apply zero product property: $x = -6$ or $x = 3$
- Domain: { x ∈ **R** | x ≠ 3 and x ≠ -6 }

10. Answer: All real numbers except x = 1, x = -2, and x = 5

- Set the denominator equal to zero: $x^3 - 4x^2 - 7x + 10 = 0$
- Test $x = 1$: $1 - 4 - 7 + 10 = 0$, so $(x - 1)$ is a factor
- Divide: $x^3 - 4x^2 - 7x + 10 = (x - 1)(x^2 - 3x - 10)$
- Factor the quadratic: $(x - 1)(x - 5)(x + 2) = 0$
- Zeros: $x = 1$, $x = 5$, $x = -2$
- Domain: all real numbers except $x = 1$, $x = 5$, and $x = -2$
- These values make the denominator zero, which makes the function undefined